

Paper A v0.6 — The Physical (SU(3))
 Yang–Mills Mass Gap via D_{IV}^5 Spectral
 Geometry. Companion to Paper B. v0.6 = v0.5
 advanced by K1708: the GENUINE spectral
 mass-gap number is $\lambda_1 = 4$ — the first
 eigenvalue of the Laplacian on the S^4 factor of
 the Shilov boundary $(S^4 \times S^1)/\mathbb{Z}_2$, tower $k(k+3)$
 $= \{0,4,10,18,28\}$, degeneracies 1,5,14,30,55, with
 the first excited level’s degeneracy exactly n_C
 $= 5$ (geometry-forced; genuinely S^4 , not S^6 —
 there is no S^6 in BST, Cal). The earlier ‘6’ is the
 CONFORMAL (SO(2)/time-circle) reading, $4 +$
 ρ -shift $2 = 6 = C_2 := n_C + 1$ — which is the
 identity Cal flagged ($C_2 \equiv$ the Bergman-rep
 Casimir $n+1$). ‘7’ is the genus, a different
 invariant. THE ONE OPEN PIECE: how much
 conformal shift sits above the 4 — Grace’s fixed
 $+2k \rightarrow 6$ vs Elie’s ν -dependent $2\nu k + k^2$
 $\rightarrow \{4,8,15\}$, 6 only at non-BST $\nu=1/2$); pinning
 the SO(2) contribution from the primary source
 is the single live task (Grace $\oplus$ Elie). The
 glueball 1720 is OUT OF THE ROOM for the
 value-pin (its $11/6$ carries $C_2=6$ — not

independent support). BANKED FLOOR: gap exists, geometry-fixed, bounded below by 4.
 Package intro HELD until the shift is reconciled. NOT a derived value above 4.
 Count HOLDS 4.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace (OS reduction + T2490/T2491 spectral casc

2026-08-19 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.6 — K1708: the genuine spectral mass-gap number is $\lambda_1 = 4$ (S^4 boundary Laplacian, degeneracy $n_C = 5$, geometry-forced); “6” is the conformal reading = the identity $C_2 = n_C + 1$; the one open piece is the $SO(2)$ conformal shift above the 4 (Grace $\oplus$ Elie); the glueball is out of the room for the value-pin; floor $\lambda_1 \geq 4$ banked; package intro HELD.

0. Two guards and one residual (read this first)

Two objects must not be fused, and one residual must not be dressed as a proof.

- Guard 1 (space). The gap we derive is the spectral gap of the Bergman–Laplacian on the bounded domain D_{IV}^5 — a *different operator on a different space* from the Yang–Mills Hamiltonian on flat $\mathbb{R}^4$. $\lambda_1 = C_2 = 6$ on our domain is not, by assertion, the $\mathbb{R}^4$ mass gap; the two are connected only through a bridge that must be exhibited, never fused (the standing K932 guard: exhibit the bridge, don’t assert a shared eigenvalue).
- Guard 2 (object). The pure-Yang–Mills (Clay) gap is the glueball (1720 MeV, 0.6% vs lattice), not the proton (938 MeV, which is the *full-QCD*, matter-included gap — a different object); we do not lend the proton’s four-decimal precision to the Yang–Mills gap (K1705). And the glueball is *out of the room* for the gap-value pin (K1708): $1720 = (11/6) \cdot 938$ carries $C_2 = 6$ inside its $11/6$ factor, so it is a real forward prediction but not independent evidence for the integer — the value is pinned from the geometry alone (the GATE-KOIDE discipline that keeps $Q = 2/3$ out of the Koide pin).
- Guard 3 (the symbol “ C_2 ” itself — K1707). Three distinct objects all read “ C_2 ”

and all equal 6 at the physical point $n = n_C = 5$, so writing “the gap = $C_2 = 6$ ” hides *which one* is meant:

- $C_2^{\text{int}} := n_C + 1 = 6$ — the *labeled integer primary* (one of the five integers). A definition.
- $C_2^{\text{flow}}(n) = 2n - 4$ — the *commitment-flow Casimir* (a function of n ; = 6 at $n=5$ only by the coincidence-at- $n_C=5$).
- $C_2^{\text{Berg}} = n + 1$ — the *Bergman-representation Casimir* (a function; = 6 at $n=5$, and equal to C_2^{int} *identically*, since C_2^{int} is *defined* as $n+1$).

The consequence (Cal, K1707): “ $\lambda_1 = C_2 = 6$ ” as previously written is C_2^{int} = C_2^{Berg} — the same definition on both sides, an identity that *cannot fail* ($P^2=P$), not a derived tower-gap. The genuine, falsifiable question is whether the *physical mass-gap operator’s* tower-gap equals 6 non-tautologically — which awaits pinning the one operator from the primary source (Section 2).

- The one residual (flat space). The honest open problem is a single question: is the theory that the reconstruction machinery (Rehren holography $\rightarrow$ Osterwalder–Schrader) places on the flat 4D boundary literally interacting SU(3) Yang–Mills, or a gapped generalized-free cousin that also satisfies the axioms? This is the large named open — carried as such, never a proof.

With those stated, the paper’s claim is explanatory, not triumphal: BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain. The gap is nonzero because the domain is bounded (real physics, Section 2), its value is geometry-fixed — no free parameter, and the genuine spectral number is $\lambda_1 = 4$ — the first eigenvalue of the S^4 boundary Laplacian, degeneracy $n_C = 5$, geometry-forced (K1708). The old “6” is that 4 dressed in the conformal ($SO(2)$) shift, which coincides with the *identity* $C_2 = n_C+1$; the *one open piece* is how much shift sits above the 4 ($\rightarrow 6$ or 8, Grace $\oplus$ Elie). The banked floor: gap exists, geometry-fixed, bounded below by 4. The flat- $\mathbb{R}^4$ construction is carried as the large named open. Nothing here is “we solved Yang–Mills,” and — after K1707/K1708 — nothing here claims a *derived* value above the genuine boundary number 4 until the conformal shift is pinned.

Abstract

We study the Yang–Mills mass gap for the physical gauge group SU(3) — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass-gap operator is the Laplacian on the S^4 factor of the Shilov boundary $\partial_S D_{IV}^5 = (S^4 \times S^1)/\mathbb{Z}_2$, and its gap is nonzero because the domain is bounded — the spectrum is discrete and bounded below, and *compactness is the gap mechanism* (decompactify and the gap slides to zero: $\lambda_1(R) \rightarrow 0$). The genuine spectral value is $\lambda_1 = 4$ (K1708): the S^4 scalar-harmonic tower is $\lambda_k = k(k+3)$, degeneracies 1, 5, 14, 30, 55, with the first level’s degeneracy exactly $n_C = 5$ — a geometry-forced computation, not an identity. The earlier “ $\lambda_1 = C_2 = 6$ ” is the *conformal ($SO(2)$)/time-circle reading*, $4 + \rho$ -shift $2 = 6 = C_2 := n_C+1$; that is an identity (C_2 equals the Bergman-representation Casimir $n+1$ by definition, so the match cannot fail, Cal K1707) — not a falsifiable derivation. The one open piece is how much conformal shift sits

above the 4: a fixed $+2k \rightarrow 6$ or a ν -dependent $2\nu k + k^2 \rightarrow \{4, 8, 15\}$; pinning the $SO(2)$ contribution from the primary source is the single live task (Grace $\oplus$ Elie, K1708). The safe, banked floor: the gap exists, is geometry-fixed with no free parameter, and is bounded below by 4. (*The glueball 1720 MeV is out of the room for this pin — its 11/6 factor contains $C_2 = 6$, so it is not independent support, K1708.*) The construction does not build 4D Yang–Mills from scratch on flat $\mathbb{R}^4$; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering) on the domain’s 4D conformal boundary ($SO(4,2) \subset SO(5,2)$), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single residual (Section 6, Section 9): is this gapped edge theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory — equivalently, is the Rehren-reconstructed boundary theory *literally* $SU(3)$ YM? In v0.4 we report that the decisive probe — the cross-channel glueball spectrum — has been computed and materially advances the interacting reading: the four ground J^{PC} channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), with one blind leg, $2^{++} = g/n_C$, and the other channels consistent within current lattice error. A generalized-free theory produces no such bound-state J^{PC} tower — real evidence on the interacting side — but the interacting-vs-free question remains the named-open residual. We do not claim the Clay problem is solved. The honest statement is a decisive *sharpening* to one residual, with the flat- $\mathbb{R}^4$ identification carried as the large named open.

1. Scope: the physical theory, not “all G”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen. So “all G” is not BST’s claim. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap: existence SOLID, genuine spectral value $\lambda_1 = 4$ (S^4 boundary Laplacian), conformal shift the one open piece (K1708)

What is SOLID (real physics). The mass-gap operator is a Laplacian on the Shilov boundary of D_{IV}^5 — a positive operator whose spectrum is the arrow of time (the heat-semigroup generator). The domain is bounded; the state space $H^2(D_{IV}^5)$ is square-integrable against the invariant Bergman measure, with the reproducing-kernel (Cauchy–Szegő / Shilov-boundary) structure fixing boundary behavior, so the spectrum is discrete and bounded below. Hence the gap is nonzero, and it is nonzero *because the domain is bounded*: decompactification ($R \rightarrow \infty$) sends $\lambda_1(R) \rightarrow 0$, killing the gap. *This compactness-is-the-gap-mechanism is the physical content, and it is not in question.* The value is geometry-fixed — no free parameter: the gap is a Laplacian eigenvalue, an integer determined entirely by the geometry, with

nothing to tune.

The genuine spectral number is $\lambda_1 = 4$ (K1708) — a real computation, not an identity. The physical mass-gap operator is the Laplacian on the S^4 factor of the Shilov boundary $\partial_S D_{IV}^5 = (S^4 \times S^1)/\mathbb{Z}_2$. Its spectrum is the S^4 scalar-harmonic tower $\lambda_k = k(k+3) = \{0, 4, 10, 18, 28, \dots\}$, with degeneracies 1, 5, 14, 30, 55, ... — a *bona fide* S^4 spectrum (not S^6 ; there is no S^6 in BST, Cal). The first excited level ($k=1$) has degeneracy exactly $n_C = 5$, geometry-forced (the S^4 vector harmonics). So the gap is $\lambda_1 - \lambda_0 = 4$ — a genuine eigenvalue of a named boundary operator, with a geometry-forced degeneracy. *This is the honest number.*

Why “6” was written, and why it was the identity (K1707/K1708). The “6” is the conformal ($SO(2)/\text{time-circle}$) reading: adding the conformal/ ρ -shift +2 to the boundary eigenvalue 4 gives $4 + 2 = 6$, coinciding with $C_2 = n_C + 1 = 6$. But that coincidence is the identity Cal flagged (C_2 defined as $n_C + 1$ = the Bergman-rep Casimir $n+1$, Guard 3), so “ $\lambda_1 = C_2 = 6$ ” was the *conformal reading equated with its own definition* — not an independent derivation. The genuine boundary computation is 4; “6” is 4 dressed in the conformal shift; “7” is the genus $g = n_C + 2$, a different invariant. Resolution of the old $\{4, 6, 7\}$:

reading	value	status
S^4 boundary Laplacian, $k(k+3)$	$\lambda_1 = 4$	genuine spectral number (K1708)
conformal reading (4 + ρ -shift 2)	6	$= C_2 = n_C + 1$ — the identity
Bergman-rep Casimir, $n+1$	6	$\equiv$ the identity (same definition)
genus / embedding invariant, g	7	a different invariant, not the gap

The one open piece — how much shift sits above the 4. Whether the physical mass-gap state carries a conformal shift, and how much, is the single unresolved question (Grace $\oplus$ Elie, K1708): is the $SO(2)/\text{time-circle}$ contribution a fixed $+2k$ ($\rightarrow \lambda_1 = 6$) or ν -dependent $2\nu k + k^2$ ($\rightarrow \{4, 8, 15\}$, with 6 only at the non-BST value $\nu = 1/2$)? The genuine boundary number 4 is banked either way; whether a 6 or an 8 sits above it rides entirely on the physical ν . Safe floor (banked): the gap exists, is geometry-fixed, and is bounded below by 4.

The glueball is out of the room (K1708) — it is NOT independent support. The glueball $1720 \text{ MeV} = (11/6) \cdot 938$ contains $C_2 = 6$ in its $11/6$ factor, so it cannot be cited as independent evidence for the gap value without circularity — the same discipline that keeps $Q = 2/3$ out of the Koide pin (GATE-KOIDE). The glueball remains a real forward prediction (Section 7), but it is exempted from the value-pin: the integer is pinned from the geometry alone. (*This corrects v0.5’s “glueball intact and separate” to “glueball intact as a prediction, but out of the room for the value-pin.”*)

Tier: gap EXISTENCE = SOLID (bounded $\Rightarrow$ gap, real physics); genuine spectral value $\lambda_1 = 4$ = SOLID (S^4 boundary Laplacian, degeneracy $n_C = 5$, geometry-forced); the conformal shift above 4 ($\rightarrow$ 6 or 8) = OPEN (Grace $\oplus$ Elie, K1708); floor $\lambda_1 \geq 4$ banked. *Guard 1 holds: a gap of a boundary Laplacian on D_{IV}^5 , not asserted to be the $\mathbb{R}^4$ YM Hamiltonian's gap.*

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only $SU(3)$ ” — is answered by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the dimension-free output of the proved T1829). $N_c = 3$ is the T1829 output (a value-recurrence, Cal #335); the $su(3)$ /color readings of “3” are *downstream of the open color identification*, not independent derivations. So $SU(3)$ is substrate-derived. The whole dynamical content cascades from one physical input — three colors fix $\text{rank} = 2$ ($N_c = \text{rank}^2 - 1$), and rank generates $\{n_C, C_2, g\}$ (T2491). The “all G ” wall is dissolved.

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data on the 4D conformal boundary

BST does not construct 4D YM from scratch on $\mathbb{R}^4$. The OS reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory carries it — on the domain's 4D conformal boundary, which exists because D_{IV}^5 carries the 4D conformal group $SO(4,2) \subset SO(5,2)$ (the same edge where light and electromagnetism live):

OS datum	D_{IV}^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID
Hamiltonian with a gap	S^4 boundary Laplacian on $H^2(D_{IV}^5)$; gap nonzero (bounded $\Rightarrow$ gap); genuine value $\lambda_1 = 4$ (degeneracy $n_C=5$); conformal shift above 4 open (K1708)	SOLID (existence + $\lambda_1=4$ floor); shift pending
reflection positivity	tube-type (Cayley $\rightarrow T(\Omega)$); Θ = cone involution; OS-RP (free level) = Cauchy–Szegő RKHS-positivity	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped	SOLID-structural

OS datum	D_IV ⁵ realization	tier
clustering / unique vacuum	from the spectral gap $\Delta > 0$ (nonzero, geometry-fixed)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. The single non-free-level piece is the interacting upgrade of reflection positivity — which is the residual (Section 6). Tier: SOLID at free level; interacting-RP = the residual.

5. Net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS (Hardy–Szegő) isometry intertwines the bulk and boundary operator nets: HS is $SO(5,2)$ -equivariant, and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (BGL) / Borchers. So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks. (*This is the modular-geometry rung of the reconstruction stack named in Section 9.*)

6. The one residual: interacting Yang–Mills, or generalized-free? (= “is the Rehren edge literally YM?”)

The prize is a single question, and F1056 and v0.3 name it two ways that are one residual:

Is the OS/Rehren-reconstructed gapped 4D boundary theory genuinely interacting $SU(3)$ Yang–Mills, or a generalized-free gapped theory?

The two namings coincide: - v0.3’s naming (interacting vs generalized-free): a generalized-free field also satisfies the OS axioms and also has a gap — so the residual is to show the reconstructed theory is genuinely *interacting*. - F1056’s naming (Rehren-edge = literally-YM): Rehren’s algebraic holography is a rigorous bulk $\leftrightarrow$ boundary net correspondence, but its *dual is not automatically the physical CFT* — so the residual is to show the Rehren edge theory is *literally* $SU(3)$ YM, not a cousin.

These are the same open piece. “Genuinely interacting” and “literally the physical YM net” are one condition, viewed from the OS side and the Rehren side respectively.

A trap we avoid. “Non-commutative $\implies$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n -point functions ($n \geq 3$). The YM self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. So

interacting $\iff$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3) \iff$ the cross-channel glueball spectrum matches $SU(3)$ (Section 7).

Two pieces of evidence toward the interacting side: - The bulk-color algebra closes as non-abelian $\text{su}(3)$ (bilinear Schwinger realization; Elie 4301). The substrate identification (#418) is in progress. - The spectrum is non-additive and computed (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state J^{PC} tower. The computed ladder is non-additive and channel-structured — evidence of binding. This is *evidence toward* interacting, not a closure of it.

7. The cross-channel glueball spectrum — named-open, materially advanced

In v0.1 this section was *named-open at structural tier*. Since v0.3 it is computed and materially advanced — still named-open (the honest tier, after Cal’s trim of the v0.2 over-claim “derived / landed interacting”).

7.1 Why there are exactly four channels (operator-algebraic closure)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{IV}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible J^{PC} components, each of which must have eigenstates for the substrate’s operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\mu\rho} F^{\nu}_{\rho})$ traceless	curvature (the stress tensor)
1^{+-}	$\text{Tr}(F[D, F])$	derivative structure

The multiplicity is forced by the tensor structure — the answer to “why four channels?” Same architectural principle as nuclear magic numbers, different driver (operator-algebra closure for the bosonic tensor vs Pauli for fermions). The odd-balls (0^{+-} , 1^{-+} , 2^{+-}) are forbidden at two gluons — clean unmixable channels.

7.2 The masses: one verified number sets the ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $\text{SO}(2)$ dilatation) on the holomorphic discrete series on $H^2(D_{IV}^5)$, diagonal in the K-type basis by Schur:

$m \propto E = \lambda_0 + (\text{energy step})$, $\lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5$ (verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$).

Energy step = $\text{SO}(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$. With one dimensionful anchor (seat = $\pi^5 \cdot m_e = 156.4 \text{ MeV}$; $m(0^{++}) = c_2 \cdot \text{seat} = 1720 \text{ MeV}$):

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev	tier
0^{++}	0	1	—	1720	1730	−0.6%	anchor
2^{++}	$J = 2$	$7/5$	g/n_C	2408	2400	+0.3%	BLIND (genus + spin)
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%	consistent (twist value- checked)
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%	consistent (twist value- checked)

Only the $2^{++} = g/n_C$ leg is fully blind (genus + spin only, nothing read from the data, lands because $g = n_C + \text{rank}$). The 0^{-+} and 1^{+-} use the half-canonical twist $n_C/2$, which is rep-motivated but value-checked against the data — so they are consistent within lattice error, not blind predictions. (Quantization note: the *ratios* live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11.)

7.3 The radial direction is linear, by Schur

The scalar holomorphic discrete series is one irreducible representation; by Schur the Casimir is constant across its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} = (1,1)$ sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~ 2670 , within quenched excited-glueball error). Consequence stated honestly: the spin-linear / radial-quadratic factorization is a clean negative — both directions are linear within the irrep.

7.4 Experimental and decay-side notes

The experimental scalar-glueball candidate $f_0(1710) \approx 1704$ MeV is consistent with BST’s 0^{++} at seat 11 (1720 MeV) — the expected lightest-glueball match (not a unique identification; the scalar sector mixes with $q\bar{q}$). The 0^{-+} coupling f_G is the substrate-computable Bergman mode norm (a Gindikin-Gamma ratio; 0^{++} kernel $= 60 = C_2 \cdot n_C \cdot \text{rank}$); with the established $\chi_{\text{top}}^{1/4} = 180$ MeV the WV identity gives $f_G(0^{-+}) \approx 12.6$ MeV. Decay-side: mixing (glueball $\leftrightarrow q\bar{q}$ overlap on the shared Hardy space), not a “dump.”

7.5 Honest disposition of Section 7

The cross-channel spectrum is computed (four channels; one blind leg $2^{++} = g/n_C$; the others consistent within lattice error; multiplicity explained; radial

direction settled by Schur; lightest state consistent with $f_0(1710)$). This materially advances the interacting reading — a generalized-free theory produces no such bound-state J^{PC} tower. It does not close the interacting-vs-free residual (Section 6). That — equivalently, the *formal interacting-RP upgrade / proving the Rehren edge is literally YM* — remains the named-open residual (Section 9).

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the glueball tower of Section 2; $g = 7$ is the $so(7)$ vector = g_2 fundamental = $3 \oplus \bar{3} \oplus 1$. Color is not a geometric isometry of the noncompact domain ($su(3)$ not-subset-of $so(5,2)$) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . The discrete-series spectrum of this $so(7)$ has the primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490). Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger realization on H^2 is in progress (#418).

9. Honest scope: the conditional, and the reconstruction stack for the one residual

We do not claim to have proved the Yang–Mills mass gap. The honest statement:

Given the reconstruction stack executes to rigor — including the interacting upgrade that identifies the edge theory as literally $SU(3)$ YM — the physical Yang–Mills theory exists as a rigorous 4D QFT with a nonzero, geometry-fixed mass gap whose genuine boundary value is $\lambda_1 = 4$ (the conformal shift above it set by the primary-source $SO(2)$ pin, K1708).

The reconstruction stack (the honest W4 machinery, named — F1056/K1705/T1271):

Hua-1963 boundary values $\rightarrow$ Bisognano–Wichmann-1975 / Borchers-2000 modular geometry $\rightarrow$ Rehren-2000 holographic net $\rightarrow$ Osterwalder–Schrader reconstruction $\rightarrow$ cluster decomposition.

Rehren’s algebraic holography is a *theorem* (rigorous, not the approximate AdS/CFT), so the residual is well-posed, not hand-waving. But it has two named, unfinished parts: (1) run the correspondence on D_{IV}^5 (build the bulk net and obtain the 4D boundary net); (2) identify the reconstructed boundary theory with the physical $SU(3)$ YM — the known subtlety that a Rehren/OS dual is not *automatically* the physical CFT. Part (2) is the genuine open piece, and it is *the same* as the interacting-vs-free residual (Section 6).

What is SOLID independent of the conditions: the gap exists and is nonzero (bounded $\Rightarrow$ gap) with a geometry-fixed value and a genuine boundary number $\lambda_1 = 4$ (S^4 Laplacian, degeneracy $n_C = 5$; the conformal shift above 4 is the one open piece, K1708 — floor $\lambda_1 \geq 4$ banked); the substrate-derivation of $SU(3)$ (Paper B); the OS *data* (Section 4); the $so(7)$ unification (Section 8). The glueball 1720

MeV at 0.6% is a real forward prediction but out of the room for the value-pin (its 11/6 carries $C_2 = 6$ — not independent support, K1708). What is materially advanced but still named-open: the cross-channel spectrum (Section 7) — real evidence on the interacting side that does not by itself close the residual. The single flat-space residual is the interacting-upgrade / literally-YM identification. This is a sharpening of the Clay problem — *not* a claim of its solution, and (after K1707/K1708) *not* a claim of a derived value above the genuine boundary number 4 until the conformal shift is pinned.

Explanatory summary (Guard 1 + Guard 2 held). *BST reproduces the scale and channel-structure of the Yang–Mills mass gap on a bounded domain*: the gap $\Delta = C_2 = 6$ is forced by representation theory (Guard 1: a domain operator, not the $\mathbb{R}^4$ Hamiltonian), its pure-glue reading matches the glueball at 0.6% (Guard 2: not the proton), and the flat- $\mathbb{R}^4$ construction is carried as the large named open — never a proof.

10. Open items and falsifiers

- The one residual (Section 6/9): the interacting-upgrade / proving the Rehren-OS edge theory is literally SU(3) YM.
- #418 substrate identification: the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) (algebra closure verified; identification in progress).
- Blind-leg audit promotion: confirm $\lambda_0 = \text{genus} = n_C$ is the Bergman lowest weight and canonical-bundle weight = genus (Elie toys A+B) — would promote the $0^{-+}/1^{+-}$ legs from value-checked to blind.
- D_A dynamical-gap check (Elie, Round 12): does the fluctuated Dirac D_A^2 carry $\lambda_1 = C_2 = 6$ dynamically (gauging preserves the winding-energy gap)? Report “the D_{IV}^5 gap,” target the glueball, not the proton.
- Falsifier: the spectrum is a forward prediction — 0^{++} degenerate with 2^{++} at 2408 MeV is testable; if the cross-channel ratios fail, the interacting reading fails (the gap value $C_2 = 6$ and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.6 (K1708): the gap exists (bounded $\Rightarrow$ gap) and the genuine spectral number is $\lambda_1 = 4$ — the first eigenvalue of the S^4 boundary Laplacian (tower $k(k+3)$; degeneracies 1,5,14,30,55; first-level degeneracy = $n_C = 5$), a geometry-forced computation, not an identity. The old “6” is that 4 dressed in the conformal ($SO(2)/\rho$) shift, which coincides with the identity $C_2 = n_C + 1$; “7” is the genus. The one open piece: how much conformal shift sits above the 4 — fixed $+2k (\rightarrow 6)$ or ν -dependent $2\nu k + k^2 (\rightarrow \{4,8,15\})$ — pinned from the $SO(2)$ /time-circle contribution (Grace $\oplus$ Elie, K1708). Banked floor: gap exists, geometry-fixed, bounded below by 4. The glueball 1720 is out of the room for the value-pin (its 11/6 carries $C_2=6$ — not independent support). Section 0 keeps the three C_2 guards. Package intro HELD until Grace $\oplus$ Elie reconcile the shift. NOT a derived value above 4, NOT “spectrum closed,” NOT “YM proved.” INTERNAL. Nothing pushed;

CP existence-only.

Draft v0.6 (K1708). Spine: Grace's S^4 degeneracy read $(1,5,14,30,55 \rightarrow \lambda_1=4$, degeneracy $n_C=5$) + OS reduction + net-compat + T2490/T2491; Elie's four-formula catch + D_A blind spectrum + the $SO(2)$ ν -structure; Lyra's operator framing + the 4-vs-conformal-6 resolution + residual unification; the $so(7)$ unification; Paper B (gauge-group derivation); Cal K1707 (identity catch) + Keeper K1708 (genuine number = 4, glueball exemption). The one open piece = the $SO(2)$ /time-circle conformal shift ($Grace \oplus Elie$). Companion to Paper B v0.6.

— Lyra, Wed 2026-08-19 (date-verified). Paper A v0.6, K1708: genuine spectral number $\lambda_1 = 4$ (S^4 boundary Laplacian, degeneracy $n_C=5$); “6” = conformal reading = the identity; glueball out of the room; floor $\lambda_1 \geq 4$ banked; package intro HELD until the conformal shift is pinned.